

Bacteriophage Therapy as a Potential Management Option for Surgical Wound Infections

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Abstract

To investigate the potential role of bacteriophages in the treatment of surgical infections, we conducted a retrospective analysis of four surgical patients who have sought treatment at the Eliava Phage Therapy Center, Tbilisi, Georgia. Two patients had chronic osteomyelitis, one presented with a diabetic foot ulcer, and the fourth patient had developed a severe infectious complication after skin grafting surgery. Patients were treated with different combinations of bacteriophage preparations, based on the sensitivity of the isolated bacterial strain toward commercially available bacteriophages. The treatment lasted on average for 1 month, and positive results were obtained in all four cases: the wounds have healed, the general health status of the patients has improved. No allergic or adverse reactions have been observed throughout the treatment.

Keywords: phage therapy, surgical infections, wound healing, antibiotic resistance, Eliava, bacteriophage

Introduction

IN THE SLOWLY approaching postantibiotic era and a time of limited therapeutic options, it has become crucial to find an alternative treatment for bacterial infections. Bacteriophages, viruses that infect bacteria, were utilized for antibacterial treatment, even before the development of antibiotic therapy; however, the discovery of penicillin in 1928 quickly replaced phage therapy and became the most widely used antibacterial in the world. Now, as bacterial resistance toward antibiotics slowly turns into a global health crisis, researchers have turned their attention back to bacteriophages and their role in the treatment of chronic infections. Especially important are surgical infections since these often become a significant cause of postsurgical complications and mortality.^{1,2}

Two scientists Frederick Twort and Felix d'Herelle independently discovered bacteriophages in the 20th century. d'Herelle was the first, who coined the term “bacteriophage,” and explicitly categorized the mysterious agent as a virus. He emphasized the potency of bacteriophage use in various pathologies and developed different complex cocktails of these viruses designed to hit different bacterial targets: “Bacte-intesti-phage,” “Bacte-staphy-phage,” etc.³

During a visit to the Pasteur Institute, Georgian scientist, George Eliava, developed a relationship with d'Herelle that

by 1930s led to the establishment of The Eliava Institute of Bacteriophages, Microbiology and Virology. Adding to the base of the already existing institute of bacteriology established in 1923. The institute has been an important scientific research institution for decades. It housed one of the biggest phage libraries and was highly active, during these decades supplying the Red Army with dysentery phages during World War II (WWI), as well as producing phage medications for hospitals throughout the Soviet Union.⁴

In 2009, Eliava scientists developed a nonprofit foundation, the Eliava Foundation, an association for the development of bacteriophages in Georgia that formed multiple spin-off companies to promote its products and services. The Eliava Phage Therapy Center (EPTC), an outpatient clinic opened in 2010, is one of the companies established by the foundation. The center is one of the few facilities in the world providing phage treatments, offering its patients a unique alternative treatment of bacterial infections by using bacteriophages. The clinic is staffed by doctors from multiple subspecialties including internal medicine, pediatrics, urology, gynecology, dermatology, and surgery.

Research on phage therapy is extensive and includes positive results both in animal research^{5–7} and in human trials.^{8–10} In countries such as Georgia, Poland, and Russia, phage therapy is practiced regularly, with an excellent safety record and has often become crucial in the treatment of cases

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wherein antibiotics have failed.^{11,12} The safety of phages has been demonstrated by empirical¹³ as well as clinical evidence.^{14,15} Rose et al.¹⁶ even argue that bacteriophages should not be considered as typical drugs and undergo the same evaluation for safety since they are “natural controllers of bacteria.”

The biological nature of bacteriophages, such as exponential reproduction at the site of infection and easy penetration when compared with antibiotics, make them favorable in the use of surgical infections.⁹ Reports of bacteriophage use in surgical patients include both historical anecdotes and case studies that date as far back as 1930. The New York Academy of Medicine published an article by Patterson and Albee,¹⁷ which described the successful use of anti-staphylococcal bacteriophage lytic for the wound strain in a 10-year-old boy diagnosed with osteomyelitis of the left tibia and humerus. It has been reported that the wound closed completely after 2 weeks of treatment.

During WWII, the Soviet army carried vials of bacteriophages as part of the routine medical supply and used it in the treatment of dysentery as well as wound infections.³ In addition to being part of first aid kits for field soldiers during WWII, phage was extensively used in battlefield wounds of the Abkhazian war of 1991, as special enhanced cocktails were found to reduce infectious complications in wounds.⁹ Decades of extensive use of phage in surgical wounds have been well documented in Georgian and Russian literature.¹⁸ Conventional treatment, including antibiotics, is crucial in the treatment of wound infections, but the problem of antibiotic resistance has created a need for alternatives, such as bacteriophage. In the following sections, we present four cases, which indicate the successful use of bacteriophages in the surgical practice at the EPTC.

Case 1

Presentation of the case

A 74-year-old male with osteomyelitis of the sternum and recurrent parasternal abscess, status postcoronary artery bypass grafting in 2011, presented at the EPTC in June of 2018. Previous treatment with antibiotics was unsuccessful, as the

infection was not resolved; subsequently the resection of the sternum was planned. At presentation, two fistulas with purulent discharge were localized on the upper third of the sternum (Fig. 1a). The diameter of the upper fistula was 0.6 cm, and that of the lower fistula was 0.4 cm. The patient's medical history is complicated with type 2 diabetes mellitus and arterial hypertension, New York Heart Association class II.

Upon presentation at the clinic, notable changes in clinical analysis included leukocytosis, increased erythrocyte sedimentation rate (ESR) (50 mm/h), increased C-reactive protein (CRP) (89.18 mg/L), and slightly increased glycosylated hemoglobin (HbA1c) (7.18%).

A thoracic computed tomography (CT) showed deformed and uneven contours of the sternum. The structure of the bone was nonhomogeneous and mostly sclerosed. Ventrally to the sternum, periosteal soft tissue was thickened with significant fibrous changes.

Staphylococcus aureus was identified as a major causative agent of the wound infection. The identified bacterial strain was tested for activity against different classes of antibiotics and commercially available bacteriophages. The identified *S. aureus* strain was sensitive toward doxycycline, ciprofloxacin, and trimethoprim–sulfamethoxazole.

Based on phage sensitivity testing, three commercial bacteriophage preparations were selected to treat the infection: staphylococcal bacteriophage, Pyo bacteriophage, and SES bacteriophage (Table 1). All bacteriophages were used *per os*, and Pyo bacteriophage was also used locally. Ten milliliter of liquid Pyo bacteriophage was sprayed on the gauze, with a sterile syringe and applied to the wound once daily. The wound was covered with sterile adhesive wound dressing.

Oral treatments with three phage preparations—staphylococcal bacteriophage (10 mL), Pyo bacteriophage (10 mL), and SES bacteriophage 10 mL—in combination after taking 100 mL of alkaline mineral water as an antacid (“Borjomi”)—were used daily. The initial phage treatment was carried out for 20 days. Antibiotic therapy was not conducted in this patient, although the patient had undergone multiple rounds of IV antibiotics in his home country without success. After a

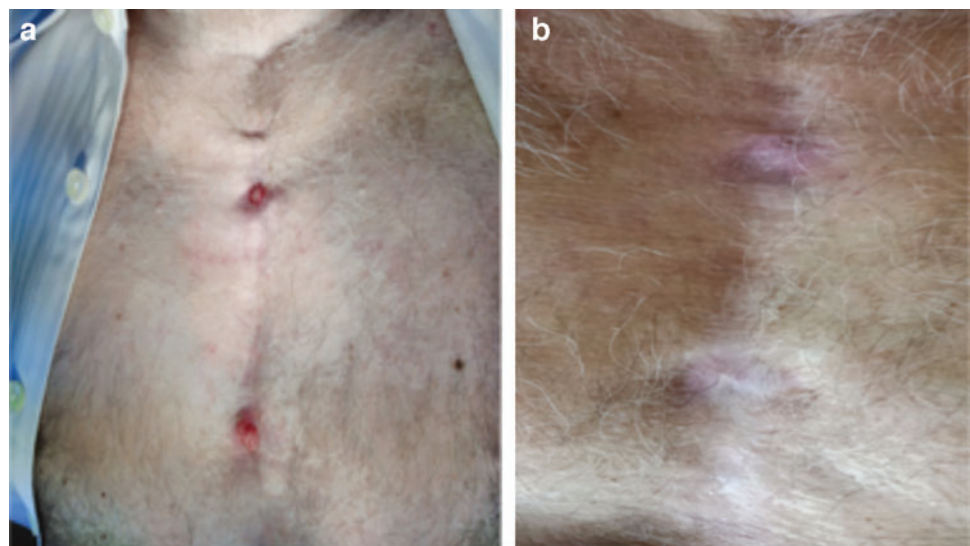


FIG. 1 (a) The image was taken on admission. Two purulent fistulas are seen in the sternal bone. The diameter of the upper fistula is 0.6 cm and the diameter of the lower fistula is 0.4 cm. (b) Postbacteriophage treatment, the fistulas have closed and there is no purulent exudate.

TABLE 1. COMPOSITION OF DIFFERENT BACTERIOPHAGE PREPARATIONS USED FOR TREATMENT

Phage preparations (produced by LTD Eliava Biopreparations)	Lytic activity against bacterial species
Staphylococcal bacteriophage	<i>Staphylococcus</i> spp.
Pyo bacteriophage	<i>Staphylococcus</i> spp., <i>Streptococcus</i> (<i>S. pyogenes</i> , <i>S. sanguis</i> , <i>S. salivarius</i> , <i>S. agalactiae</i>), <i>Escherichia coli</i> , <i>Pseudomonas aeruginosa</i> , <i>Proteus</i> (<i>mirabilis</i> and <i>vulgaris</i>).
SES bacteriophage	<i>Staphylococcus</i> spp., <i>Streptococcus</i> spp., and <i>Enteropathogenic E. coli</i> of different types.
Intesti bacteriophage	<i>Shigella flexneri</i> (Serotypes 1,2,3,4); <i>Shigella sonnei</i> ; <i>Shigella newcastle</i> ; <i>Salmonella paratyphi</i> A, <i>Salmonella paratyphi</i> B; <i>Salmonella typhimurium</i> ; <i>Salmonella enteritidis</i> ; <i>Salmonella choleraesuis</i> ; <i>Salmonella oranienburg</i> ; <i>E. coli</i> ; <i>Proteus</i> (<i>vulgaris</i> , <i>mirabilis</i>); <i>Staphylococcus aureus</i> ; <i>P. aeruginosa</i> ; <i>Enterococcus faecalis</i> .

2-week phage-free period, the treatment was resumed for an additional 2 weeks, using the scheme already presented.

Results

Discharge from the fistula started to reduce after day 3 of therapy, granulation tissue began to form from day 7. The wounds were completely covered with granulation tissue within the first 20 days of treatment. After the second round of the phage treatment, the diameter of the wound decreased and the purulent fistulas closed (Fig. 1b). No adverse effects were observed. Observation after treatment for 1 year did not show exacerbation of osteomyelitis.

Table 1 lists the composition of different bacteriophage preparations. The staphylococcal bacteriophage contains only one type of phage. The other three preparations (Pyo bacteriophage, SES bacteriophage, and Intesti bacteriophage) are mixtures of phages against different pathogens. According to the standards established by the Ministry of Health of Georgia, the titer of individual phage components in the phage preparations not $>10^7$ plaque forming unit per milliliter of solution (pfu/mL). Preparations are mixture of sterile filtrates of phage lysates. Staphylococcal bacteriophage is highly purified (by column chromatography), and the other three preparations are purified by filter sterilization.

Case 2

Presentation of the case

A 60-year-old female presented with chronic osteomyelitis of the right tibia after a hospital-acquired *S. aureus* infection 30 years ago. Since the accident, the patient has suffered from chronic pain. In 2005, a large fistula developed in the right tibia. A year later, the injured area had undergone necrosis requiring operative treatment, wherein antibiotic cement was placed during reconstructive surgery. In August 2016, three ulcers developed in the injured area, and the patient came to the EPTC for treatment. On admission, the patient experienced intense pain of the leg, and mobility was severely restricted. The patient's medical history includes type 2 diabetes mellitus, diagnosed in 2010. Blood analyses showed increased HbA1c (8.4%). Culturing of wound sample revealed *S. aureus*.

Upon physical examination, the skin appeared red, very delicate, and fragile. Purulent ulcers were visible on the surface of the skin: the middle ulcer was 2.5×1.5 cm and the upper one was 0.5×0.5 cm in dimension (Fig. 2a,b).

Sensitivity of the isolated bacterial strain—*S. aureus*—was tested against antibiotics and was found to maintain sensitivity to most groups of antibiotics. Based on the phage sensitivity testing, five of the commercial phage cocktails were found to be active against the isolate. Two bacteriophage preparations were selected to treat the infection: staphylococcus bacteriophage and Intesti bacteriophage (Table 1).

Both phages were used *per os*, and staphylococcus bacteriophage was used externally. Ten milliliter of liquid staphylococcus bacteriophage was sprayed on to the gauze with a sterile syringe until the gauze was soaked and applied directly onto the wound. The wound was covered with sterile adhesive wound dressing. Oral treatment with two phage preparations—staphylococcus bacteriophage 10 mL once daily and Intesti bacteriophage 10 mL once daily—was also taken *per os* after using 100 mL of alkaline mineral water (“Borjomi”) as an antacid.

The initial phage treatment was carried out for 20 days. The patient took a 3-week break and continued the treatment *per os*: staphylococcus bacteriophage 10 mL once daily in the morning, Pyo bacteriophage 10 mL once daily in the afternoon, and SES bacteriophage once daily in the evening, for 15 days. Pyo bacteriophage ointment was also applied twice daily for 15 days. *Staphylococcus aureus* was highly sensitive to all four bacteriophages, and all four phage cocktails were used at different stages of treatment. No antibiotic treatment was carried out during phage therapy.

After 3 weeks of a phage-free period, based on the results of repeated laboratory testing, the patient continued treatment with Pyo bacteriophage 10 mL once daily in the morning and SES bacteriophage in the evening *per os* in combination with the Pyo bacteriophage ointment, for 15 days. Repeat bacteriology was not performed because no discharge was observed to sample, the wound had already started to dry out and close.

One year later, the patient carried out an oral prophylactic phage treatment course, despite her satisfactory condition including stable remission of osteomyelitis.

Results

The patient's condition improved after phage treatment. Visually, the ulcers filled in, the fibrotic-purulent layer cleared, and granulation tissue started to form (Fig. 2c). Throughout the treatment, the patient did not experience any allergic reactions or adverse effects. The patient had a

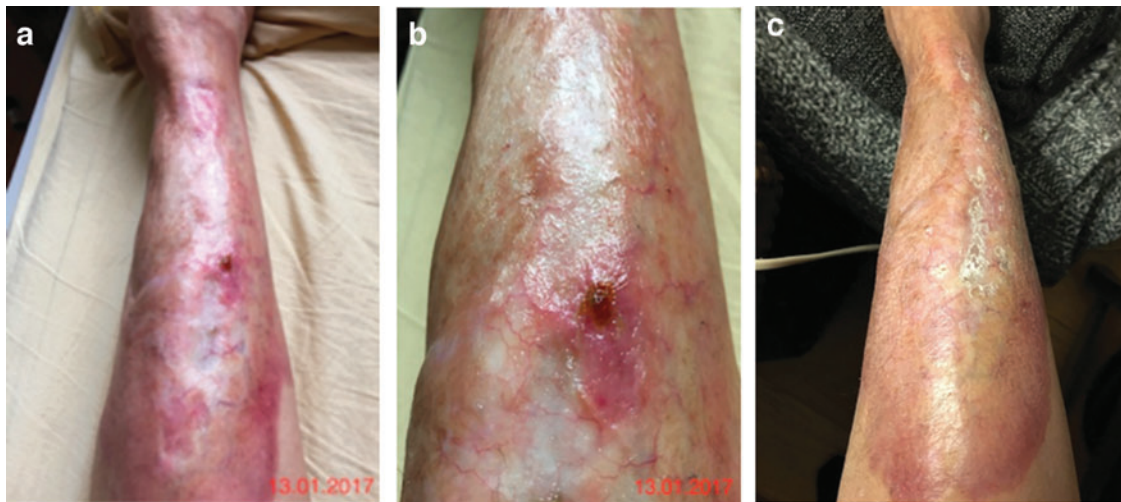


FIG. 2 (a) Pretreatment, the skin on the bone is markedly red, delicate, and fragile. (b) Purulent ulcer is visible. (c) Post-treatment, the color of the skin has changed from red to pink. The ulcer has filled in completely.

complete closure of the wounds at 18 weeks of treatment, and 2 years post-therapy, no recurrence of osteomyelitis has been observed.

Case 3

Presentation of the case

A 69-year-old male with antibiotic-resistant bacterial infection presented at the EPTC in July of 2017. The patient had type 2 diabetes mellitus as a complication of which,

several months prior, he developed a diabetic foot ulcer on the left foot (Fig. 3a). The patient underwent long-term treatment with antibiotics, which was ineffective. Complete blood count (CBC) showed increased ESR (55 mm/h), increased HbA1c (7.6%), and decreased international normalized ratio (1.78). Creatinine (208.46 mmol/L) and uric acid levels (268.46 mmol/L) were both elevated. Bacterial culture of samples taken from the ulcer was polymicrobial and showed *Burkholderia cepacia*, *S. aureus*, and *Enterococcus faecalis* growth, with *S. aureus* identified as the



FIG. 3 (a) Before treatment, on the left toe, diabetic foot ulcer is seen. The wound is 5 cm in diameter. (b) Catheter is placed into the ulcer for direct application of phages into the tissue. (c) Six weeks after treatment; the size of the wound reduced to 2 cm by 1.8 cm. (d) Post-bacteriophage therapy, the ulcer has filled in completely and the wound has closed.

major causative agent. X-ray of the foot showed changes typical for osteomyelitis. During a physical examination, the wound was ~5 cm in dimension, with visible purulent exudate.

All three strains isolated from the wound were tested for antibiotic sensitivity and were seen to be multidrug resistant. Based on the bacterial sensitivity, two bacteriophage preparations were selected to treat the infection: staphylococcus bacteriophage and Intesti bacteriophage. A phage specifically against *B. cepacia* was not available. Both phage preparations were used *per os*, and Intesti bacteriophage was also used externally. Ten milliliter of liquid Intesti bacteriophage was sprayed on with a sterile syringe to soak the gauze, which was applied to the wound, once daily. In addition, a sterile catheter was inserted into the ulcer for local irrigation of the infected tissue with 2–3 mL Intesti bacteriophage once daily for 7 days (Fig. 3b).

Oral treatments with two phage preparations, staphylococcus bacteriophage 20 mL and Intesti bacteriophage 20 mL, were taken after 100 mL of alkaline mineral water (“Borjomi”) was taken as antacid. The initial phage treatment was carried out for 20 days, after which the patient took a 2-week pause and continued the same treatment for another 20 days. Other medications used were B vitamin complex (Milgamma100 mg) and oral filtered extract of calf blood as a supplement (Actovegin 200 mg). The patient did not undergo antibiotic therapy at any time during phage therapy.

Results

Five weeks post-treatment, the size of the wound reduced to 2.4 cm by 2.5 cm. After 6 weeks, the area was further reduced to 2 cm (Fig. 3c). Visually, the ulcer healed and closed, the fibrotic-purulent layer resolved, and granulation tissue has started to form. Signs of osteomyelitis on CT have also resolved. No adverse events to phage preparations were observed throughout the treatment. The patient did not have recurrence after 1 year of treatment (Fig. 3d).

Case 4

Presentation of the case

A 68-year-old male presented at the EPTC in December of 2017 with postsurgical infection. The patient underwent skin graft surgery from the front surface of the left thigh to cover a surgical wound left after the removal of the larynx due to carcinoma. After the skin graft surgery, the area where the skin had been removed became infected, which, despite antibiotic treatment, did not heal (Fig. 4a). Upon admission, CBC was performed: CBC, ESR, and CRP were within the normal range, but HbA1c was above the reference range (5.96%). Bacterial culture of swabs taken from the infected area showed *Pseudomonas aeruginosa*. Ultrasonography of the subcutaneous tissue showed subcutaneous infiltration, but no subcutaneous abscess was noted. Antibiotic sensitivity of the isolated *P. aeruginosa* strain showed multidrug resistance.

Based on the bacterial sensitivity, two bacteriophage preparations were selected to treat the infection: Pyo bacteriophage and Intesti bacteriophage. Both phage cocktails were used *per os*, and Pyo bacteriophage was also used locally. Ten milliliter of liquid Pyo bacteriophage was sprayed

on with a sterile syringe to soak gauze, which was applied to the wound, once daily. The wound was covered with sterile adhesive wound dressing.

Oral treatment with two phage preparations—Pyo bacteriophage 10 mL once daily and Intesti bacteriophage 10 mL once daily—was also taken after neutralizing the stomach acid with 100 mL of “Borjomi” alkaline mineral water. The initial phage treatment was carried out for 20 days, after which the patient took a 2-week pause and continued the same treatment for another 20 days. Solcoseryl, an epithelial regenerative powder, was also used once daily, in the treatment course of this patient. After initial treatment, the tissue started to regenerate, the redness of the skin subsided, and granulation tissue formed. Antibiotics were not used in combination with phage therapy.

In May of 2019, the patient addressed EPTC, to seek treatment of a second skin graft surgery wound on the front surface of the right thigh. Despite initial intravenous (IV) and local antibiotic therapy in his home country, the infection persisted, the patient applied to our clinic after 1 month postsurgery (Fig. 4b). Bacteriological analysis revealed infections caused by *S. aureus* and *Serratia marcescens*. Treatment with bacteriophage preparations was initiated, following the same protocol as described earlier. Since no commercially available bacteriophage preparations include a phage active to *S. marcescens* species, and preparation of an individual phage would not be timely, we decided to use phages against the dominant pathogen *S. aureus*.

Results

After 3 months of phage treatment, the wound on the left thigh has healed completely (Fig. 4c). The same successful results were obtained for the right thigh wound treatment as well: the infection was resolved, and the tissue has healed completely (Fig. 4d). During both stages of treatment, the patient has not experienced any adverse effects during or after phage therapy.

Discussion

The EPTC has treated a number of patients with surgical wounds including osteomyelitis, prosthetic-associated infections, abscesses, and fistulas. Most of the wounds treated are caused by multidrug-resistant bacterial strains. Treatment of surgical infections with phages is a well-established practice in Georgia since the physicians and military authorities actively explored the therapeutic potential of bacteriophages in surgical practice starting from 1930s to 1940s. During the Finnish War, a Georgian scientist Alexander Tsulukidze, who has previously worked on the development of staphylococcus and streptococcus phages with Eliava and d’Herelle, was allowed to test the efficacy of phages in the Leningrad Red Army Hospital.¹⁹ Tsulukidze has reported that during a 5–6-day treatment, the improvement was usually seen after the first 2 days of phage application.¹⁸

In the former Soviet Union, phages were used for multiple purposes: treatment of purulent wounds, management of postoperative infections, and prevention of wound infections.²⁰ The route of phage administration varied from oral to intravenous to topical, and the highest success rates were reported in cases of *staphylococcal* and *streptococcal* infections.²⁰

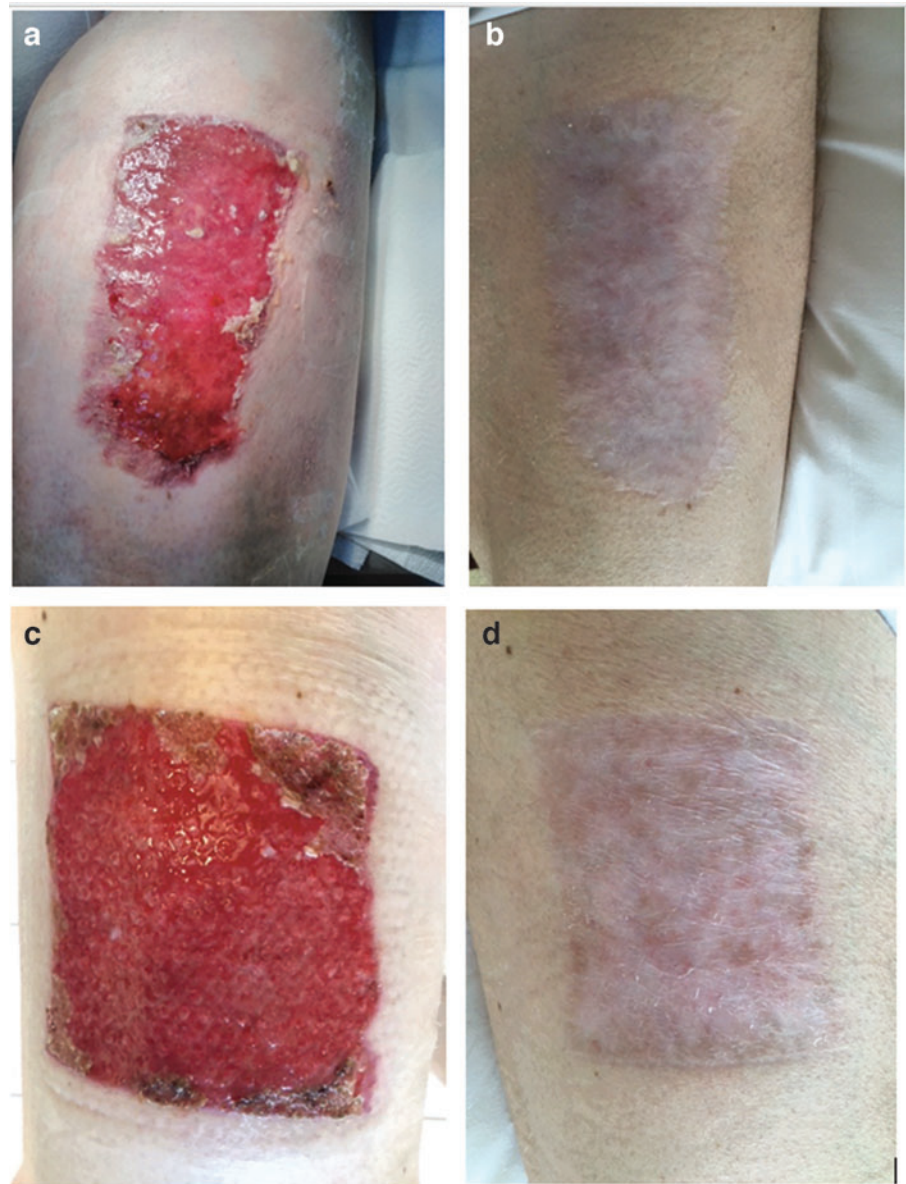


FIG. 4 (a) Pretreatment, on the upper half of the left thigh, there is a 15×25 cm superficial bleeding wound. (b) Postbacteriophage treatment, the wound on the left thigh has closed completely. (c) Pretreatment, on the upper right thigh: 15×25 cm superficial bleeding wound. (d) Postbacteriophage treatment, the wound on the right thigh has closed completely.

This report presents a case series of four patients with chronic recalcitrant bacterial infections of various etiologies. Two of the patients presented with chronic osteomyelitis, one patient with infection of the sternum and recurrent parasternal abscess and the other with infection of the right tibia. Both cases with osteomyelitis were caused by MSSA *S. aureus* infections. The third patient developed a diabetic foot ulcer. The bacteriological analysis of swabs taken from the ulcers showed polymicrobial infection of *B. cepacia*, *S. aureus*, and *E. faecalis*. The fourth patient had a *Pseudomonas* infection during the first visit for a postoperative wound, and during the second visit, a mixed infection with *S. marcescens* and *S. aureus*.

Before coming to the EPTC, all four patients had undergone prolonged antibiotic therapy, which was ineffective. At the EPTC, each bacteriophage treatment regimen was selected individually. The isolated bacterial strains were tested for sensitivity against commercially available bacteriophage preparations. Postbacteriophage treatment, improvement and wound healing were seen in all four cases. Importantly, no

adverse reactions to phage treatment were observed throughout the treatment.

In all the four cases discussed, *S. aureus* was identified as the major causative agent. This pathogen is the most common cause of bone infections in all age groups.⁵ The use of antibiotics in the treatment of osteomyelitis is complicated not only because of possible antibiotic resistance but also because antibiotics often demonstrate poor and inadequate penetration to such infection sites.⁸ The positive effect of bacteriophages in the treatment of osteomyelitis has been documented in numerous animal research studies. Kishor et al. have successfully used bacteriophages isolated from water sources to treat chronic osteomyelitis in rabbits.⁶ After 2 weeks of therapy, the rabbits recovered from the infection. Their activity and appetite improved, and their erythema, induration, and local edema decreased.⁶

Ibrahim et al also used rabbits to demonstrate that bacteriophage treatment was effective in treating multidrug-resistant *S. aureus* chronic osteomyelitis.⁵

Diabetic neuropathy is a leading problem in health care; it is estimated that out of 350 million patients diagnosed with diabetes, 30–50% will develop peripheral neuropathy, eventually resulting in limb amputation.²¹ The treatment and management of diabetic foot ulcer includes surgical debridement, management of infection with antibiotics, and revascularization.²² In addition to the problem of resistance, antibiotics are easily metabolized by the human body, and they often do not reach the target tissue in therapeutically significant doses.²³ Since the blood supply is severely compromised in the extremities of diabetic patients, it is more difficult for orally absorbed medications to reach the site of infection.⁸

The pharmacokinetics of phage application is drastically different from antibiotics and offers several advantages, including exponential growth, easy penetrance to the target sites, specific host range, and minimal toxic effects. Owing to their longevity and evolution, phages could be superior to antibiotics in some cases, especially when antibiotic resistance is a problem.^{21,24} Fish et al.²⁵ used the staphylococcal phages produced by the Eliava Biopreparations in a case series of six patients with diabetic foot ulcers and observed recovery of the underlying infection as well as the complete healing of the ulcers for a 7-week period.

Since the route of admission can affect the disease progression, it is important to consider the pharmacokinetics of bacteriophages. It has been shown that different modes of administration are preferable in different cases and with different pathogens.²⁶ *Staphylococcus aureus* associated biofilms have been proven to pose a serious challenge to wound healing, including immunogenic properties, which hinder immune response to infection.²⁷ Seth et al.⁷ demonstrated that the combined use of tissue debridement and bacteriophages reduced bacterial load and improved healing parameters in *S. aureus* biofilm-infected wounds. During the treatment of the diabetic foot ulcer, the surgeon in this study decided to introduce phages directly into the tissue by using a sterile plastic microcatheter, in combination with topical and oral application of phages.

This case was especially complicated because the infection was caused by three different pathogens: *B. cepacia*, *S. aureus*, and *E. faecalis*. However, as a result of the bacteriophage treatment, the ulcer closed completely. The phage used did not target *B. cepacia*; however, we observed eradication of all bacterial strains. The same phenomenon was seen with the mixed infection caused by *S. marcescens* and *S. aureus*. This concept has been referred to by other authors as the “head of the snake” theory. The theory hypothesizes that the underlying root of the infection is *S. aureus*.²⁵

Staphylococcal phage has also been successfully used in the eradication of *S. aureus* osteomyelitis in different case studies. Fish et al.²⁸ recently reported complete eradication of *S. aureus* osteomyelitis in the distal toe phalanx of a 63-year-old diabetic patient and no recurrence 3 years post-treatment. The metagenomic analysis of microbial ecology in wounds may lend further insight into the pathophysiology of chronic nonhealing wounds.^{27,29,30}

In case 1 reviewed in this report, the patient with *S. aureus* osteomyelitis of the sternal bone experienced a significant change in antibiotic resistance during the bacteriophage treatment. The sensitivity of the identified *S. aureus* strain toward antibiotics was performed twice: during the first test, resistance was seen toward 12 antibiotics, whereas during the second test, which was performed 13 days post-treatment,

resistance was seen only toward 7 antibiotics. During the treatment, the patient was given a combination of staphylococcus bacteriophage, Pyo bacteriophage, and SES bacteriophage. The mechanism behind change in resistance has been explained by Buckling and Rainey,³¹ who suggest that as part of the directional coevolution, bacteria undergo mutational change in receptor sites causing the loss of virulence.

This observation has been supported by *in vitro* studies as well. Zhang and Buckling³² concluded that combined use of kanamycin and phages decreases the chance of resistance evolution of *Pseudomonas fluorescens*. According to Ryan et al.,³³ use of cefotaxime and T4 bacteriophage for the eradication of bacterial biofilms *in vitro* is more successful than the use of antibiotics alone. Positive results have also been obtained with the combined use of bacteriophages and ceftolozane/tazobactam *in vivo* studies of joint and bone infections caused by *P. aeruginosa*.³⁴ In addition, Chhibber et al.²¹ demonstrated that the combined use of lytic bacteriophage and linezolid reduced bacterial load, lesion score, and other histopathological parameters in diabetic foot infections caused by *S. aureus*.

Evaluation of these four surgical cases, presented at the EPTC, once again confirms the long history of beneficial use of bacteriophages. Continued use for surgical infections in Georgia shows that phages can be successfully used in the treatment of bacterial infections of various pathologies. Further exploration into microbial wound ecology may provide insight into wound care. Phage therapy offers a viable alternative for the treatment of antibiotic-resistant infections in patients who have a high risk of complications of surgical wounds. The potential of this approach must be studied as a prospective controlled study to better assess efficacy of bacteriophage application in infected surgical wounds.

Authors' Contributions

L.N., N.H., D.N., and N.P. were involved in patient management. N.B. was involved in bacteriology and bacteriophage laboratory investigations. N.N. and N.H. drafted the initial report. L.N., M.Z., and E.K. reviewed and edited the article. All authors read and approved the final article.

Author Disclosure Statement

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